

First-half Lab Series — ten weeks, ten artefacts

Each handout maps to a 90-minute Übung session. W08 carries graded weight.

TEN LAB SHEETS

W01 Doc-Types Ethnography Identify 4 doc types in a chosen repo. Cite URLs. Fr 18:00	W02 IEEE 830 vs 29148 Compare two SRS standards across 6 axes. Fr 18:00	W03 Rinzler Method Story → FR-12 (60–90 min). 1 page. Fr 18:00	W04 arc42 Bausteine Map M7 to 12 arc42 sections. 1 page. Fr 18:00	W05 OpenAPI 3.0 Mini-spec: request, error, schema. 1 page. Fr 18:00
W06 Docstrings + Ethics Audit 3 docstrings; cite 1 AI antipattern. Fr 18:00	W07 Microcopy Repair 3 error messages from M7. 1 page. Fr 18:00	W08 Midterm Practice Quiz + SRS analysis + 3-min pitch. Graded. Date fixed	W09 M3 DoD Update Extend your M3 SRS with 4 DoD criteria. Fr 18:00	W10 mkdocs CI Setup Vale + lychee + GitHub Actions. 1 page. Fr 18:00
Format PDF (printable) or hand-written scan. Both count.		Submission LMS upload by Friday 18:00 each week. Late = 0.5× / day.		Grading Pass / Conditional / Fail. W08 is graded; others are go/no-go for M-milestones.

Submission window per week

W01	Doc Types Ethnography	Fr 18:00
W02	IEEE 830 vs 29148 Vergleich	Fr 18:00
W03	Rinzler Method FR–12	Fr 18:00
W04	arc42 Bausteine Map	Fr 18:00
W05	OpenAPI 3.0 Mini-Spec	Fr 18:00
W06	Docstring Audit + Ethics	Fr 18:00
W07	Microcopy Repair	Fr 18:00
W08	Midterm Practice	Date fixed
W09	M3-DoD Update	
W10	mkdocs+vale+lychee CI	

Doc Types Ethnography — 4 types per real repo

Pick one open-source repo. Find one example per type. Cite URL for each.

90-MINUTE AGENDA

00–10 min	Pick a repo Open-Source, >100 stars, >5 contributors. Avoid toy repos.
10–45 min	Find 1 example per type User / Developer / Operations / Process. Cite URL of each.
45–75 min	Note 1 anti-pattern Per type: which anti-pattern (Tribal-Knowledge, Hero, Cargo-Cult) is visible?
75–90 min	Submit 1 page Table: 4 types x {URL, anti-pattern, why}. Hand-scan or PDF.

DELIVERABLE

1 page — four-row table: Type · URL · Anti-pattern · Reason.

Due Friday 18:00 via LMS.

FOUR TYPES — CLASSIFY, THEN FIND

TYPE 01 User docs Tutorials, READMEs, quickstarts for end-users.	TYPE 02 Developer docs API references, architecture, ADRs.	TYPE 03 Operations docs Runbooks, incident responses, deployment guides.	TYPE 04 Process docs ADRs, decision logs, retrospectives.
--	--	--	---

Hinweis. „Tribal Knowledge” findet ihr in Operations-Dokumentation am häufigsten — Suche nach Phrasen wie „ask Ali” oder „see Albus” im Repo.

IEEE 830 vs ISO 29148 — six axes, one decision

Pick six axes from the W02 lecture. Compare both standards. Mark which one wins per axis.

90-MINUTE AGENDA

00–10 min	Pick six axes Ch. 2 der beiden Standards strukturiert vergleichen. Mind. 3 achsenkonkrete Details.
10–55 min	Compare axis-by-axis Note 830 vs 29148. Welcher Standard ist konkreter pro Achse?
55–80 min	Decision: M3 SRS Template Welche Achse gewinnt für eure M3-Arbeit? Begründung 1 Satz.
80–90 min	Submit 1 page Tabelle mit Achsen, je 2 Spalten Standards + Winner.

DELIVERABLE

1 page — six-axis comparison table. Plus 1 decision sentence.

Due Friday 18:00 via LMS.

SIX AXES — EXAMPLE TEMPLATE

Axis	IEEE 830	ISO 29148	Winner (for M3)
Functional reqs	5.3.1 numbered	6.4.1 stakeholder-tagged	...
Verifiability	implicit	explicit clause	...
Traceability	13.b implied	9.5 explicit matrix	...
Stakeholder tags	absent	present	...
Prioritization	implicit	muss/soll/kann	...
Architecture	separate SDD	embedded	...

Hinweis. „Winner“ ist nicht absolut — pro Achse. Für M3 (Studi-bachelor mit 15 FR) ist Verifiability wichtiger als Stakeholder-Tagging. Wählt das ehrlich.

Rinzler Method — from story to FR-12

Take a user story. Rewrite it as a testable functional requirement.

90-MINUTE AGENDA

00–10 min	Pick a story Aus eurem M7-Library oder eine neue. Schwach formuliert.
10–45 min	Rewrite as FR-N ID · source story · priority (muss/soll/kann) · verification · owner.
45–75 min	Test-of-Failure Schreibt einen CRASH-Test: 5 s Timeout, 20 MB Limit, HTTP 413 wenn exceeded.
75–90 min	Submit FR-12 Auf LMS. Pair-Audit mit Sitznachbar:in (5 min).

DELIVERABLE

1 testable FR with ID, source-story, priority, verification, owner.

Due Friday 18:00 via LMS. Pass rule: all 5 fields filled.

WORKED EXAMPLE — WEAK STORY VS FR

Weak user story

„The system should handle big PDF files gracefully.” – Alice (Backend Lead, W03 dailystandup)

Testable FR

FR-12 · Upload must accept PDF ≤ 20 MB
Source: Alice W03 story-3 · Priority: **muss**
Verification: 1 s response at 20 MB, 5 s timeout, HTTP 413 on overflow
Owner: **A. Müller** (Backend)

Hinweis. Wenn eure Verifikation „manuell testen“ ist, ist das —1/5. Test-of-Failure MUSS eine Maschine messen können: Response-Time, Bytes, HTTP-Statuscode.

arc42 Bausteine Map — 12 sections per project

For each of the 12 arc42 sections, name what your project puts there — or marks „N/A”.

90-MINUTE AGENDA

00–05 min	Pick the project Euer M7-Library oder eine echte Arc42-Repo (z. B. https://arc42.org/template).
05–60 min	Fill 12 rows Pro Baustein: Name eures Inhalts + URL/Page-Reference.
60–80 min	Note gaps Welche Bausteine sind leer? Welche sind zu fett?
80–90 min	Submit 1 page 12-Row table + 1 gap-summary sentence.

DELIVERABLE

1 page — 12-row table (Section · Content · URL/Ref). Plus gap-summary.

Due Friday 18:00 via LMS.

TWELVE BAUSTEINE — REFERENCE TEMPLATE

01 Introduction & Goals	02 Architecture Constraints	03 System Scope & Context	04 Solution Strategy
05 Building-Block View	06 Runtime View	07 Deployment View	08 Cross-cutting
09 Architecture Decisions	10 Quality Requirements	11 Risks & Debt	12 Glossary

Hinweis. Leere Bausteine sind eine Aussage: euer Projekt hat sie nicht. Schönt euch, eine leere Zeile einzureichen — Tutor entscheidet, ob das eine Disziplin oder eine Lüge ist.

OpenAPI 3.0 Mini-Spec — one endpoint, complete

Pick one FR from your M3 SRS. Write a complete OpenAPI spec for the corresponding endpoint.

90-MINUTE AGENDA

00–05 min

Pick one FR

Pick the FR you wrote in W03.

05–30 min

Write request spec

Body, params, headers, content-type.

30–55 min

Write response spec

200 + 4xx + 5xx. JSend shape or schema.

55–80 min

Add example bodies

Request example, response example, error example.

80–90 min

Submit + lint

Submit YAML. Run spectral-lint or redocly if available.

DELIVERABLE

1 endpoint fully specified: request, response, examples. OpenAPI 3.0 YAML.

Due Friday 18:00 via LMS.

WORKED EXAMPLE — FR-12 (FROM W03)

```
# FR-12 /upload endpoint – OpenAPI 3.0

paths:
  /upload:
    post:
      summary: Upload PDF ≤ 20 MB
      requestBody: multipart/form-data, PDF
      responses:
        200: { status: ok, file_id: uuid }
        413: { error: payload_too_large }
        500: { error: internal_error }
```

Hinweis. Wenn 200 keine schema-Definition hat, ist das —2/5. Schema-first-Denken ist die Kerndisziplin der API-Doku: jede Antwort braucht eine Form, sonst ist es nur ein HTTP-Status.

Docstrings + Ethics — audit three, cite one antipattern

Three docstrings from your M7 library. One AI-ethics anti-pattern named.

90-MINUTE AGENDA

00–05 min	Pick three docstrings From your M7-Library. One obvious good, one obvious bad, one ambiguous.
05–40 min	Audit each Roles: Argüment-Doku · Return-Typ · Beispiel · Edge-Cases. Score 0–5 pro Achse.
40–65 min	Name one AI antipattern Tatsächlich gefunden in eurem Repo: TODO-from-LLM / Lying-Comment / ...
65–80 min	Rewrite one bad docstring Antipattern ausbessern. Generator-konform (PEP 257 / NumPy / Google).
80–90 min	Submit audit table 1 page with 3 rows + 1 AI antipattern name.

DELIVERABLE

1 page — audit table (3 docstrings, axes 0–5), 1 AI antipattern name.

Due Friday 18:00 via LMS.

THREE AI ANTI-PATTERNS TO RECOGNISE

ANTI 01 TODO-from-LLM Generierte Doc enthält „TODO“-Style-Placeholder. Wirkt ehrlich, ist aber Verlegenheits-Luecke. Fix: TODO raus, oder explizit „NOT IMPLEMENTED“.	ANTI 02 Lying-Comment Doc beschreibt Funktion, die etwas anderes tut. Klassisch nach Rename ohne Doc-Update. Fix: rename triggert Doc-Rewrite als CI-Hook (W10).	ANTI 03 Over-confident Spec Generierte Doc macht Versprechen, die Implementation nicht hält. „Always“ für eine Funktion mit exception. Fix: „Raises ValueError when input is None“ explizit.
--	---	---

Hinweis. Erwischt ihr keine AI-Anti-Patterns in eurem Repo, sucht im M7-Library-Issues-Tracker. Falls auch dort nichts: das ist ein Befund (und ehrenwert 1 Satz in der Hand-in).

Microcopy Repair — three error messages from M7

Apply the W07 microcopy rules. Old version + new version + reasoning per repair.

90-MINUTE AGENDA

00–10 min	Pick three error messages From your M7-Library (errors.py / toast-notifications).
10–30 min	Audit each Use-Words? Consequence-named? Recovery-action? Edge-cases?
30–65 min	Rewrite New text ≤ 30 words per message. Rule 5: no jargon.
65–80 min	Peer review 5 min per pair. Strength · Risk.
80–90 min	Submit repair table 1 page with 3 rows (old · new · reason).

DELIVERABLE

1 page — 3-row repair table (old · new · reason).

Due Friday 18:00 via LMS.

FIVE MICROCOPY RULES — REFERENCE

R1 Use „We“ Self-referential: „We could not parse your request“.	R2 Consequence+Recovery Always: what happened + what to do.	R3 Empty State 1 sentence + 1 action. „No notifications yet“.	R4 Onboarding = 1 Job One step per screen.	R5 No Jargon Always use plain words. Avoid internal terms.
--	---	---	--	--

Hinweis. arc42 § 12 Glossary ist Pflicht für Rule 5: jeder Begriff, der nicht selbsterklärend ist, gehört ins Glossar. Sonst ist eure Microcopy kein Repair, sondern Verlegenheits-Luecke.

Midterm Practice — Quiz + SRS Analysis + Pitch

Three parts. Total 30 %. This lab is the rehearsal before the live midterm.

90-minute agenda — rehearsal for the live midterm

00–20 min	Part 1 — Theory quiz (8 questions) Old past-quiz. 2 min/question. Bhatti Ch. 2/6/9, Chinchilla Ch. 2/9.
20–30 min	Walkthrough + scoring Discuss 2 highest-scoring items. 1 min/Q.
30–60 min	Part 2 — SRS analysis 30 min for 2 quizzes or one extended SRS analysis.
60–80 min	Part 3 — Pitch rehearsal Three pitches · 3 min each. Live SRS-Reference required.
80–90 min	Reflection + self-grade Note 1 Strength, 1 Risk. Honour what passed and what failed.

DELIVERABLE

Three parts:

Quiz-score sheet · SRS-Analyse-Notiz · Pitch-Recording-Aufgabe.

Datum: **W08 Mittwoch** — gepflegt. Erscheinen ist Pflicht (M3).

MIDTERM RULES · ONE-SHOT GRADING

Pflicht: erscheinen, SRS-Material zum Lab mitbringen (Print oder Live-URL). **Mindest-Punktzahl:** 50 % Total (Quiz 30 % + SRS 40 % + Pitch 30 %). **M3-Meilenstein:** erreicht bei ≥ 50 %. Darunter: einmal korrigieren möglich, sonst Studium-Abbruch in der Vorlesung.

Hinweis: W08 ist graded. Bringt eure SRS live mit, vorbereitet auf drei Minuten-Pitch. Quiz dauert 20 Min, danach SRS-Analyse 30 Min, Pitches abschließend.

M3-DoD Update + Just-Enough Heuristic

Extend your M3 SRS with the four W09 DoD criteria. Apply one just-enough heuristic.

90-MINUTE AGENDA

00–10 min	Re-read M3 SRS Find the 15 FR rows. Mark: which ones meet the four W09 DoD criteria already?
10–45 min	Extend with 4 DoD columns Reader-first · Live-URL · CI-green · KI-verified. Per FR row.
45–75 min	Apply one heuristic Cut-to-ship: keep 80 % of value, drop 20 %. Document 2 cuts with reason.
75–90 min	Submit updated SRS 2-pages. Old SRS-row + new columns + 2 cuts + reasons.

DELIVERABLE

2 pages — M3 SRS-rows with 4 DoD columns + 2 cuts with reasons.

Due Friday 18:00 via LMS.

FOUR M3-DOD CRITERIA — ADD AS NEW COLUMNS

DOD 01 Reader-first Auf welcher FR-Antwort liest der Maintainer? Welcher Leser profitiert?	DOD 02 Live-URL URL der Demo. HTTP 200. Nicht „coming soon“.	DOD 03 CI-green Vale, lychee, mkdocs strict alle grün. PR rot bis sie pass.	DOD 04 KI-verified LLM-Output als Vorschlag, nicht als Wahrheit. Cite source.
--	--	---	---

Hinweis. Wenn mehr als 30 % eurer FR-Rows ein DoD nicht erfüllen, ist das ein Sprint-Stop-Signal. Schreibt 1 Satz: „W10 Tooling-Pipeline wird 4 FR nicht grün bekommen, weil ...“. Das ist eine self-discipline.

mkdocs + Vale + lychee CI — pipeline build

Build the four-step pipeline locally. Push to a branch. Confirm one merge gate.

90-MINUTE AGENDA

00–10 min	Install four tools mkdocs / Material theme / Vale / lychee. Per Pip/Cargo.
10–35 min	Configure mkdocs.yml strict: true, Material theme, 2-column nav.
35–55 min	Configure .vale.ini Styles path, MinAlertLevel=error, ReadMe rule.
55–75 min	Add GitHub Actions .github/workflows/docs.yml— 4 steps, running on PR.
75–90 min	First PR red → green Push a doc-PR. Fix until green.

DELIVERABLE

PR — first green. .github/workflows/docs.yml + .vale.ini + one passing doc change.

Due Friday 18:00 via LMS.

FOUR-STEP PIPELINE — THE DELIVERABLE SHAPE

```
1  $ mkdocs build --strict
2  $ vale --config=.vale.ini docs/
3  $ lychee --offline --no-progress ./site
4  GH PR merge if all green
```

Hinweis. Wenn eure CI-Hook lychee nicht in den Modus *--offline* geht, ist das —3/5. Lychee muss ohne Internet laufen können, sonst ist eure Pipeline nicht reproduzierbar.

Ten hand-ins, three milestones, one M3

Late = 0.5×/day. W08 is graded. M3 is the gate.

SUBMISSION CALENDAR

Week	Handout	Length · format	Deadline
W01	Doc Types Ethnography	1 page · table	Fr 18:00
W02	IEEE 830 vs 29148	1 page · table	Fr 18:00
W03	Rinzler FR-12	1 FR row	Fr 18:00
W04	arc42 Bausteine	1 page · 12 rows	Fr 18:00
W05	OpenAPI 3.0	1 endpoint · YAML	Fr 18:00
W06	Docstring Audit	1 page · table	Fr 18:00
W07	Microcopy Repair	1 page · 3 rows	Fr 18:00
W08 GRADED	Midterm Practice	Quiz+Analysis+Pitch	Date fixed
W09	M3 DoD Update	2 pages	Fr 18:00
THREE MILESTONES — ONE M-GATE	THREE MILESTONES — ONE M-GATE	THREE MILESTONES — ONE M-GATE	THREE MILESTONES — ONE M-GATE

M3 SRS Lieferung Erreicht nach: W08 Midterm (Pass) + W09 DoD-Update. ≥ 50 % W08 + 4 DoD-Spalten	M5 Architecture freeze Erreicht nach: W04 arc42 + W05 OpenAPI. 12 arc42 + 1 endpoint spec	M7 API Library live Erreicht nach: W06 + W07 + W10 CI-Build. 3 audits + 3 repairs + CI green
---	---	--

Submission policy · W01–W10

- LMS upload only. Email = not accepted. Cloud links (Drive, Dropbox) need a stable URL.
- Late = 0.5× / day. After 7 days = forfeit. W08: keine Wiederholung. M3-Tor: ein Conditional-Patch möglich.
- Format — PDF, scan, photo, YAML, PR-Link. Markdown not accepted for submissions.
- Eigenleistung — Plagiarismus führt zu Sofort-Fail, insbesondere in W07 Microcopy + W08 Midterm.